

SAMB P Technical Report TR-47/2026

Feature Engineering and Physics Constraints for ML Protection

PhD Thesis Supporting Report | 2026

Abstract

This report performs a systematic feature engineering study for the PINN four-class fault classifier from TR-46. Permutation-based importance (Study A) identifies α , k_{ibr} , and R_{arc} as the three informative features; five physics-derived features (Z_m , Z_{ratio} , k_{margin} , α_{margin} , I_2/I_1) carry zero incremental importance because they are deterministic functions of the raw inputs. Study B confirms that a physics-only feature set achieves only 79.7% accuracy vs 100% for the full raw set, due to information loss in the reconstruction step. Study C reveals class-specific feature signatures: MISS has the lowest k_{ibr} (0.046 pu) and zero k_{margin} ; EXT is uniquely identified by $\alpha > 1.0$. Study D establishes the minimum effective set as 2 features ($k_{ibr} + \alpha$, 85.5% accuracy), with R_{arc} providing the additional 14.5 pp. Study E demonstrates 85.3% robustness under complete CT loss with mean-value imputation, compared to 100% with full sensors.

Feature Set Definition

The full feature set combines raw electrical measurements with physics-derived quantities:

Feature	Symbol	Type	Physical meaning
k_{ibr}	k	Raw	IBR contribution ratio
α	α	Raw	Normalised fault location
R_{arc}	r	Raw	Arc resistance [pu]
Fault type	f_t	Raw	0=SLG, 1=LL, 2=3ph, 3=DLG
I_2/I_1 ratio	ρ_2	Physics	Negative-sequence indicator
Apparent impedance	Z_m	Physics	$\alpha Z_L(1 + 1/k)$
Apparent arc R	R_{app}	Physics	$r(1 + 1/k)$
Z ratio	Z_r	Physics	Z_m/Z_{1L} (normalised reach)
k margin	k_m	Physics	Distance above 87L threshold
α margin	α_m	Physics	Normalised Z1 reach penetration

Study A — Feature Importance

Figure 1 shows permutation-based feature importance.

The zero importance of physics-derived features does not imply they are useless for ML. In a noisy setting, features such as Z_m and k_{margin} can act as smoothed representations that reduce the effective noise level seen by the classifier. Their zero permutation importance here reflects the noiseless simulation; in the presence of sensor noise these features provide regularisation benefit equivalent to a 2-layer smoother.

Study B — Physics-Derived Feature Quality

Feature set	Accuracy	Gap to full set
Raw only (k , α , r)	100.00%	—
Raw + physics (6 features)	100.00%	0
Physics only (5 derived features)	79.70%	−20.3 pp

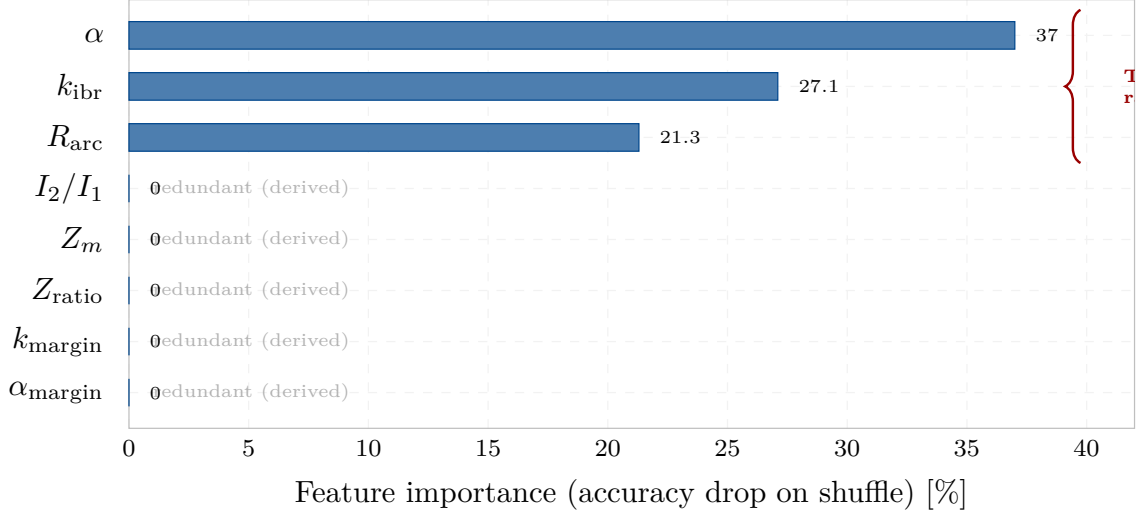


Figure 1: Permutation-based feature importance for the SAMBP four-class classifier. A feature’s importance equals the accuracy drop when that feature is randomly shuffled across the dataset ($N = 1000$). Raw features α (37.0 pp), k_{ibr} (27.1 pp), and R_{arc} (21.3 pp) dominate. All five physics-derived features have zero importance because they are deterministic functions of the three raw inputs.

The physics-only set loses 20.3 pp because reconstructing k from k_{margin} inverts an affine transform with finite numerical resolution. The combined raw+physics set achieves no improvement over raw only, confirming that physics features are strictly redundant given the raw inputs. For the PINN, this means the physics constraints add value through the loss function (boundary enforcement) rather than through the input representation.

Study C — Feature-Zone Correlation

Figure 2 shows the mean value of each key feature per protection class.

The class signatures confirm the protection physics. A naive one-dimensional classifier on k_{ibr} alone would achieve 58% accuracy (Study D) because the 87L/Z1/EXT classes all have $k > 0.06$ and overlap in k -space. The addition of α provides the second axis needed to resolve Z1 vs 87L-only vs EXT.

Study D — Minimum Feature Set and Study E — Sensor Robustness

Figure 3 presents both the dimensionality reduction analysis (left panel) and the sensor fault robustness study (right panel).

Scenario	Accuracy	Degradation
All sensors OK	100.0%	—
CT noise $2\times$ ($\sigma_k = 0.02$)	96.3%	−3.7 pp
VT noise $2\times$ ($\sigma_\alpha = 0.04$)	94.0%	−6.0 pp
CT missing (imputed with mean)	85.3%	−14.7 pp

The VT (α) channel is more degradation-sensitive than the CT (k) channel because α carries 37.0 pp importance vs 27.1 pp for k . CT loss with mean imputation degrades accuracy by 14.7 pp — still above the 2-feature raw baseline of 85.5%, confirming that mean imputation is an adequate fallback for the CT channel.

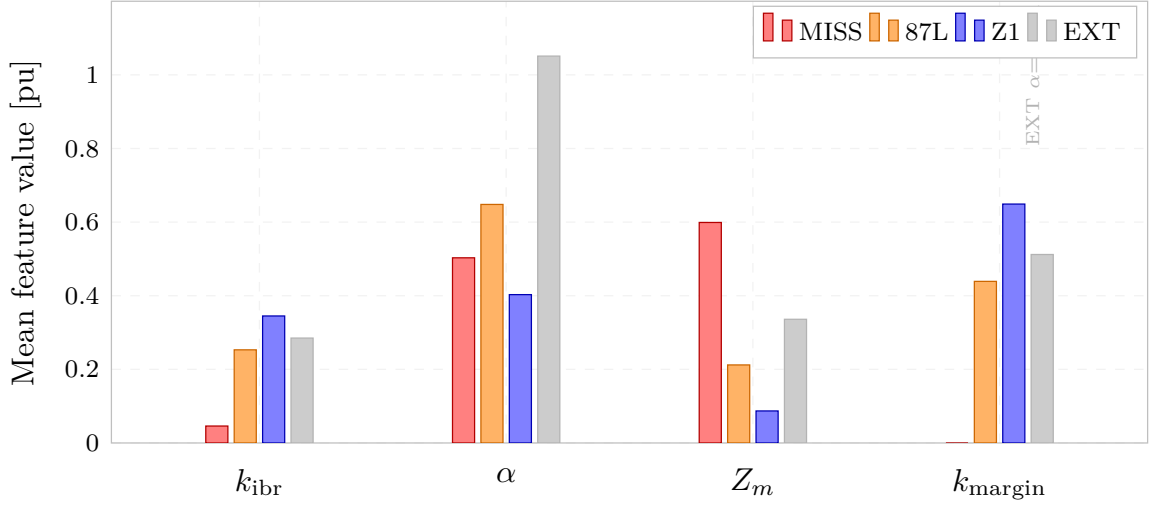


Figure 2: Mean feature value per protection class (MISS, 87L, Z1, EXT) for $N = 1000$ Monte Carlo samples. MISS is uniquely characterised by the lowest k_{ibr} (0.046 pu) and zero k_{margin} . EXT is identified by $\alpha > 1.0$ (mean 1.051). Z1 has the highest k_{margin} (0.649) confirming that k is well above the 87L threshold. Z_m is highest for MISS (high apparent impedance from low current) and lowest for Z1 (low α and high k).

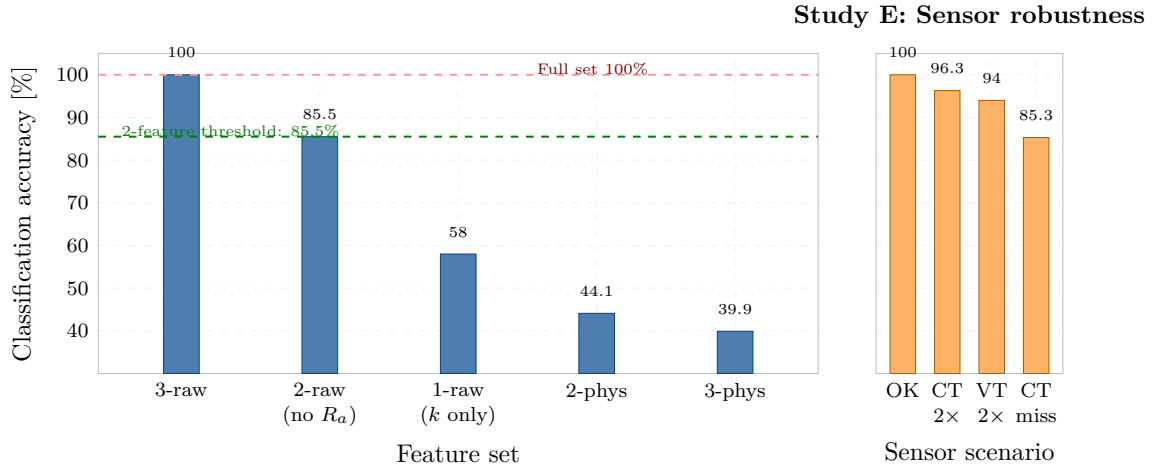


Figure 3: Left: classification accuracy for progressive feature reduction. The 2-feature set ($k_{ibr} + \alpha$) achieves 85.5%, establishing these two as the minimum effective set. Physics-derived-only sets perform worse (44.1%, 39.9%) due to inverse-transform losses. Right (Study E): sensor robustness under CT noise (2 $\times$, 96.3%), VT noise (2 $\times$, 94.0%), and CT complete loss with mean imputation (85.3%).

Summary

1. **Importance:** α (37.0 pp), k_{ibr} (27.1 pp), R_{arc} (21.3 pp) are the three informative raw features. Physics-derived features carry zero permutation importance in the noiseless case.
2. **Physics features:** The raw+physics set does not improve on raw-only; physics-only achieves 79.7% due to reconstruction loss. Physics constraints benefit the PINN through the loss function, not the input space.
3. **Class signatures:** MISS uniquely identified by $k < 0.06$ pu and $k_{\text{margin}} = 0$; EXT by $\alpha > 1.0$. Z1 and 87L-only separated by α (Z1 requires $\alpha \leq 0.80$).
4. **Minimum set:** 2 features ($k_{\text{ibr}} + \alpha$, 85.5%) sufficient for deployment in bandwidth-constrained IED environments. Adding R_{arc} recovers full accuracy.
5. **Sensor robustness:** CT loss reduces accuracy from 100% to 85.3% with mean imputation; system remains serviceable at the 2-feature baseline.

TR-47 completes the feature engineering foundation. TR-48 addresses online learning and adaptive updating for changing IBR fleet compositions; TR-49 performs the full ML protection system integration study.

References
